

Letter of Recommendation for Xuan Thanh Trinh

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Viet Nam, July 22, 2026

To: Ph.D. Selection Committee

Doctoral Program in Electrical, Electronics and Computer Engineering

College of Engineering and Computer Science (CECS)

University of Michigan–Dearborn, MI 48128, USA

senior researcher / he is yellow

Dear Admissions Committee,

I am ~~Dr.~~ Tran Thanh Son, a Lecturer in the School of Electrical and Electronic Engineering at Hanoi University of Science and Technology (HUST). As an expert in power system optimization and microgrid architectures (Ph.D., Shibaura Institute of Technology), I served as Xuan Thanh Trinh's direct research mentor on a complex project regarding peer-to-peer energy trading frameworks. Based on his exceptional performance, I firmly advocate for his admission to the Ph.D. program at the University of Michigan–Dearborn ✓

Thanh demonstrates a rare capacity for mathematical abstraction and independent research execution. ✓ Rather than passively applying formulas, he rigorously evaluates their fundamental analytical properties before deployment. Notably, he independently established the global convergence proof for an ADMM framework applied to a Second-Order Cone Programming (SOCP) relaxation, which effectively convexifies complex, non-linear grid constraints. As an independent researcher, he consistently navigates dense academic literature to master theoretical frameworks outside the undergraduate curriculum, requiring minimal direct supervision. This dual capability, profound mathematical rigor coupled with exceptional self-direction, proves his readiness for independent Ph.D. research.

Furthermore, Thanh exhibits exceptional network perspective and structural validation methodology. Specifically, he rejected the standard simplification of microgrids as passive loads and directly modeled the complex, bidirectional energy exchanges between autonomous microgrids and the distribution grid under high renewable penetrations to capture true network dynamics. To verify the robustness of his framework, he demonstrated critical methodological rigor when he challenged his system formulations with severe stress testing across complex radial and meshed topologies, including the IEEE 123-bus benchmark, completely avoiding simplified test cases that only ensure easy convergence. His ability to bridge abstract mathematical models with realistic grid constraints provided the foundational computational evidence for our manuscript, currently under review at Sustainable Energy, Grids and Networks (SEGAN). ✓

Ultimately, Thanh's methodological resilience sets him apart during complex simulations. When debugging large-scale grid networks, he constructively treats algorithmic non-convergence as diagnostic data rather than failures. Even under strict deadlines, he systematically traces numerical instabilities to their theoretical root causes rather than relying on unverified heuristics. This analytical endurance proves he is exceptionally well-equipped for the rigor of a Ph.D. program.

In summary, Thanh uniquely combines mathematical rigor, network perspective, and methodological resilience. As his mentor and co-author, I give him my highest recommendation for admission to your Ph.D. program without reservation. I am confident he will be an immediate and highly productive contributor to your research group. Please contact me if you require further information. ✓

Sincerely,

Dr. Tran Thanh Son

Lecturer, School of Electrical and Electronic Engineering

Hanoi University of Science and Technology (HUST), Vietnam

